Start with what we can observe

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Observation / ES: observación / PT: observação

Something we record or measure, like a position at a time.

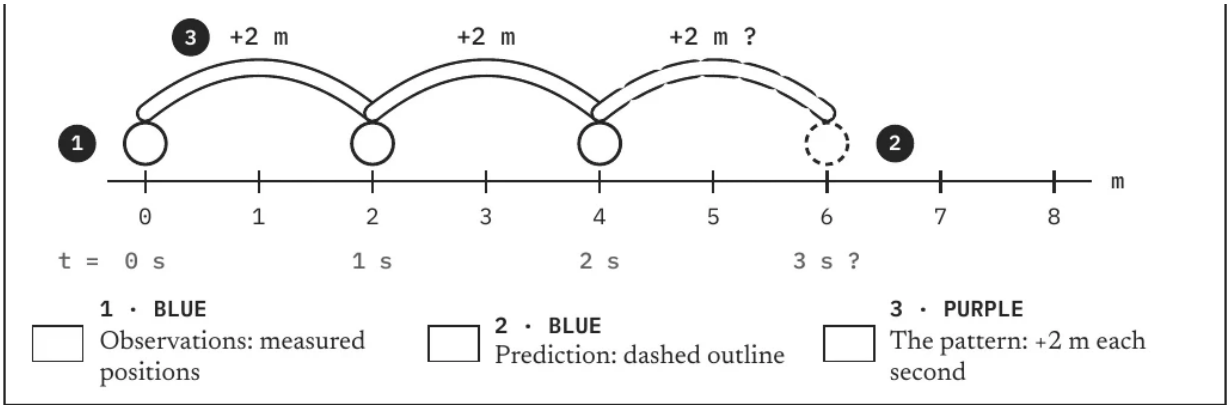
Also on your note page: Model, Prediction, Testable question.

B READ THE RELATIONSHIP

An observation is something you see or measure. A prediction says what you expect later.

Position tells where. Velocity tells how fast and which way.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart is at 2 m, 3 m, then 4 m, one second apart. Predict its position one second later.

Given x = 2, 3, 4 m 1 s between readings

Rule next = last + change each second

Work change = 3 − 2 = +1 m next = 4 m + 1 m

Answer 5 m if the motion continues the same way

E SAVE FOR THE FINAL MEETING

The desk has a sighting, not a forecast. Everything we learn in Unit 1 serves one question: how fast is 2026-XJ moving, and where will it arrive?

Read the physics: Observation

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

A telescope records where an object is at a known time. That is an observation. A future dot is a prediction, even if it follows a clear pattern.

02 HOW IT WORKS

Give each observed dot a time. Compare the gaps between dots. Equal gaps in equal times suggest constant velocity. Draw a dashed dot for the next position. It is still a guess based on a pattern.

03 WHAT TO CHECK

If a new dot misses the predicted position, check the measurement and the motion model.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart is at 2 m, 3 m, then 4 m, one second apart. Predict its position one second later. 5 m if the motion continues the same way.

05 CONNECT TO XJ

The desk has a sighting, not a forecast. Everything we learn in Unit 1 serves one question: how fast is 2026-XJ moving, and where will it arrive?

SAY IT BEFORE YOU WRITE

Which dot did someone measure? The measured dot is ___. The predicted dot is ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Observation / ES: observación / PT: observação

Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning To me, observation means …	**How can you tell?** What did someone actually see or measure? Someone saw or measured ___.
Example or sketch I see it when …	**Non-example** This looks close, but …

B DRAW WITH YOUR TEACHER

Draw three observed positions and one predicted position. Build a key that separates the two.

Labels you can use: measured · predicted.

Choose zero, direction, and scale.

Tell a partner: The measured dot is ___. The predicted dot is ___.

On assignment: Observation

SECTION E / BACK / FINAL MEETING ONLY Name _____

Three telescope fixes are at 2, 4 and 6 km at minutes 0, 1 and 2. The night crew has one more exposure.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Telescope scheduler**

Mark where to aim at minute 3. Tell the operator which dots were measured and which dot is your prediction.

☐ **Science broadcaster**

Give a two-sentence update to viewers. Point to a measured dot and say what the camera should see next if the pattern holds.

Word bank: measured · predicted.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”